

Sujan Parajuli

Northfield, VT — (802) 224-5437 — sparajul@norwich.edu
[linkedin.com/in/sujan-parajuli-618478260](https://www.linkedin.com/in/sujan-parajuli-618478260) — github.com/sujanparajuli

Research Profile: Mathematics and Computer Science undergraduate interested in machine learning, statistics, mathematical modeling, and computational finance. Research experience spans graph theory, large-scale healthcare prediction, agricultural data analysis, and ongoing mathematical and machine-learning studies of the Nepal Stock Exchange. Preparing for Ph.D. study involving data-driven mathematical research.

Education

Norwich University, Northfield, VT

Expected May 2027

B.S. in Computer Science and Mathematics, Dual Major

GPA: 3.90/4.00

Minors: Actuarial Science, Information Assurance

Relevant Coursework: Data Structures, Algorithms, Machine Learning, Database Systems, Computer Architecture, Probability, Statistics, Regression Analysis, Linear Algebra, Discrete Mathematics, Calculus 1–3

Research and Technical Skills

Programming and Reproducibility: Python, C++, R, SQL, LaTeX, Git, GitHub, Jupyter Notebook, Linux/Unix

Statistical and Machine Learning Methods: Regression, classification, cross-validation, feature engineering, time-series analysis, statistical modeling, model evaluation

Libraries and Data Tools: Pandas, NumPy, SciPy, Scikit-learn, TensorFlow, XGBoost, Excel

Systems: MSP430 Assembly, embedded C, Verilog, computer architecture, Wireshark

Research Experience

Mathematics Research

Mathematics Capstone Researcher – NEPSE Mathematical Modeling

Aug 2026 – Present

Norwich University, Northfield, VT — Mentor: Dr. Sean Kramer

- Investigated NEPSE market predictability through return analysis, correlation, and regression, establishing a statistical framework for interpreting stock-price behavior.
- Evaluated forecast reliability through baseline comparisons and performance analysis across time periods, identifying modest predictive gains that varied over time.
- Developing a 12-page capstone paper synthesizing mathematical methods, empirical evidence, and limitations to assess the scope of NEPSE predictability; expected at completion.

Weintz Undergraduate Research Fellow – Vertex Magic Trees

May 2025 – Jul 2025

Norwich University, Northfield, VT — Mentor: Dr. Dan McQuillan

- Investigated vertex-magic total labelings through combinatorial arguments and computational searches, producing examples and testing structural bounds for tree families.
- Constructed and verified tree labelings using algorithms and averaging arguments, identifying a counterexample with 5 internal vertices and 8 leaves.
- Synthesized 20+ scholarly sources through literature review and computational analysis, communicating mathematical findings in a research poster and technical report.

Computer Science and Data Research

Computer Science Capstone Researcher – NEPSE Forecasting

Aug 2026 – Present

Norwich University, Northfield, VT — Mentor: Prof. Charles Snow

- Built a reproducible NEPSE forecasting pipeline using Python, Jupyter, and chronological validation, connecting historical market data with next-trading-day prediction.
- Developing stock-price and direction models using regression, classification, and LSTM methods, with baseline comparisons to assess performance on unseen trading dates.
- Preparing a 12-page capstone paper documenting system design, model evaluation, and forecasting limitations to communicate reproducible results; expected at completion.

Undergraduate Researcher – Healthcare Predictive Modeling

Jan 2025 – Apr 2025

Norwich University, Northfield, VT — Mentor: Dr. Natalie Cartwright

- Analyzed 10,000,000 HCUP NEDS 2016 records using demographic, insurance, location, income, and diagnosis variables to investigate emergency-department charge patterns.
- Built an XGBoost workflow using 10,000-row data chunks, invalid-record filtering, and categorical encoding, enabling model training within available computing resources.
- Evaluated predictions using an 80/20 training/test split and error metrics, obtaining a test R-squared of 0.28 and identifying limitations in diagnosis grouping.

Poster Presentations

VMTL and Its Properties

Dec 2025

Norwich University Poster Showcase — Mentor: Dr. Dan McQuillan

Predicting Emergency Department Costs Using Demographic and Socioeconomic Factors: A Preliminary Machine Learning Approach

Apr 2025

Norwich University Poster Showcase — Mentor: Dr. Natalie Cartwright

Teaching Experience

Peer Tutor – Mathematics and Statistics

Aug 2023 – Present

Norwich University, Northfield, VT

- Tutored 20+ students through 1-on-1 guided practice in calculus, statistics, and pre-calculus, helping strengthen quantitative reasoning and independent problem solving.
- Staffed tutoring hours for 2 professors using concept review and worked examples in Calculus 1, Calculus 2, and Elementary Statistics, supporting assignment and exam preparation.

Professional Experience

Arena IT Helper

2026 – Present

Kreitzberg Arena, Norwich University, Northfield, VT

- Designed 5+ advertisements using arena-specific visual content, supporting hockey and season-pass promotions.
- Supported ticket sales through Purplepass, scheduling coordination, and content updates, helping maintain arena event and technology operations.

Conference Services Team Leader

Summer 2026

Norwich University, Northfield, VT

- Coordinated summer programs serving 300+ students through housing plans, access arrangements, and event schedules, supporting arrivals, departures, and facility transitions.
- Resolved operational issues through coordination with camp directors and campus service teams, keeping room assignments and event information current.

Farm Data and IT Assistant

Summer 2025, Summer 2026

Green Mountain Girls Farm, Northfield, VT

- Cleaned and organized soil data by structuring soil-test results, creating records for comparing nutrient conditions across farm areas.
- Analyzed soil macronutrients using statistical and machine-learning methods, identifying spatial patterns to support farm-management decisions.
- Supported farm events through volunteering and assistance with activities, helping event operations run smoothly.
- Maintained the farm website through content updates and IT troubleshooting, supporting daily operations.

Student Supervisor

Sep 2023 – May 2025

Sodexo, Norwich University, Northfield, VT

- Trained 5+ student employees through hands-on instruction, daily assignments, and shift coordination, maintaining coverage during high-volume service.

Leadership and Activities

Vice President, Billiards Club

Jan 2025 – Present

- Coordinate a 20-player club through practices and tournaments, supporting member participation.

President, Nepali Student Union

Aug 2024 – Present

- Lead the executive board through budgets and campus partnerships, delivering cultural and community events.

Orientation Leader

Fall 2025

- Guided 43 incoming students through orientation activities and campus-resource introductions, helping them navigate university schedules and expectations.

Junior Civilian Ring Committee Member

2025

- Coordinated Junior Ring ceremony logistics through collaboration with vendors and university staff, helping deliver an event for 200+ attendees.

Certifications and Languages

Certifications: Microsoft and LinkedIn Career Essentials in Data Analysis and Cybersecurity

Languages: English (Fluent), Nepali (Native), Hindi (Conversational)